

LoopyPro MIDI Control Reference

Condensed reference for controlling LoopyPro from external MIDI controllers (Midifighter Twister, webapp via iConnectivity AUDIO4c). Covers the binding model, all available actions, and the specific CC mapping used in the explorer-ps / MC6 Project setup.

Current Setup (Explorer)

Hardware path: Webapp → WebMIDI → iConnectivity AUDIO4c USB1 → iPad (LoopyPro)

Channel: 16 (configurable, stored in localStorage)

CC Mapping:

CC	Action	LoopyPro Identifier	Subject
20	Record	Track Record/Stop	None (global)
21	Play/Stop	Track Play/Stop	Selected Track
22	Clear	Clear Track	Selected Track
23	Mute	Mute	Selected Track
24	Solo	Track Solo	Selected Track
25	Multiply (×2)	Multiply Track	Selected Track
26	Divide (÷2)	Divide Track	Selected Track
27	Previous (◀)	Track Select	Global (param: Previous)
28	Next (▶)	Track Select	Global (param: Next)
30–39	Select loop 0–9	Track Select	Position index

Track layout: 5 color groups × 2 tracks = 10 tracks total. Colors: Orange, Yellow, Lime, Blue, Magenta. Each group has an A track and a B track.

Message format: Momentary CC — send value 127, wait 50ms, send value 0.

Project generation: `node music/MC6 Project/explorer/generate-lpproj.js` produces an `.lpproj` bundle with all bindings pre-configured. AirDrop to iPad.

Binding Model

Trigger → Binding → Action(s)

- **Trigger:** Any incoming MIDI message (CC, Note On/Off, Program Change)
- **Binding:** Maps a trigger to one or more chained actions

- **Profile:** Container for bindings. Can be **Project** (stored in `.lpproj`, can reference specific clips) or **Global** (available across all projects, should not reference specific clips)

Gesture Differentiation

A single MIDI trigger can be split into three gestures:

Gesture	Detection
Tap	Single press/release
Hold	Sustained >500ms
Double-Tap	Two presses within 250ms

Key parameters:

- **Defer Other Actions:** On Hold/Double-Tap bindings, prevents the Tap action from firing prematurely
- **Preempt Other Actions:** On Record/Play actions (default ON), fires on single tap even when double-tap is configured. Often needs disabling.

Action Chaining Timing

Timing	Meaning
With Last	Simultaneous with previous action
After Last	After previous action completes
Next Trigger	On next press of same trigger (multi-step button)
Quantized	At next clock-aligned moment
Delay	After specified time (supports seconds.ms in 2.0+)

MIDI Learn

1. Enter MIDI Learn mode (main menu)
2. Tap on-screen element to bind
3. Send MIDI message from controller
4. Binding is created in the chosen profile
5. Edit in Control Settings to add more actions, change timing, etc.

Complete Action Catalogue

Clip Actions

Action	Description
Play/Stop	Toggle playback. Option: "Record If Empty"

Action	Description
Record	Record/stop recording. Options: "If Clip Has Audio" → Overdub / Play If Stopped / Overdub If Playing. "After Recording" → Play / Stop / Overdub
Solo	Solo clip (mutes all others)
Mute	Mute/unmute
Clear Clip	Erase audio
Adjust Parameter	Continuous: Volume, Balance (pan), Pitch, Speed
Adjust Playhead	Scrub/jump playback position
Multiply Length	Double clip by repeating contents
Divide Length	Halve clip
Reverse	Toggle reverse playback
Peel/Replace Layers	Undo/redo overdub layers
Select	Mark as selected (white dot)
Phase Align	Align clip phase with master clock
Cancel Count Ins/Outs	Cancel pending operations

Color Group Actions

Action	Description
Play/Stop	All clips of a color
Solo / Mixer Solo	Solo a color group
Mute	Mute a color group
Adjust Parameter	Color group Volume, Balance

Clock Actions

Action	Description
Toggle Clock Pause	Play/pause master clock
Tap Tempo	Tap tempo input

Action	Description
Adjust Tempo	Continuous tempo control (jog)
Adjust Beats Per Bar	Change time signature numerator
Set Master Length	Change master cycle length (affects quantization)
Toggle Metronome	On/off
Adjust Metronome Volume	Continuous
Reset Clock	Reset tempo to unset
Toggle MIDI Clock Sync	Enable/disable MIDI clock
Toggle Ableton Link	Enable/disable Link sync

Effect Actions

Action	Description
Enable/Disable Effect	Toggle on/off
Adjust Effect Parameter	Continuous control of any AUv3 parameter
Select Effect Preset	Switch presets

Session Actions

Action	Description
Undo / Redo	Session-level undo/redo
Adjust Master Volume	Continuous
Cancel Pending Actions	Cancel all quantized actions
Load Project	Set list navigation
Save Project	Save
Toggle Sequence	Start/stop sequencer
Toggle Mixer	Show/hide mixer

Controller Actions

Action	Description
Switch Controller Profile	Change active profile via MIDI
Trigger Widget	Trigger on-screen widget (macro buttons)
Switch Page	Navigate canvas pages
Send MIDI Message	Send arbitrary MIDI to a destination

Follow Actions (Event-Driven)

Trigger automatically on state changes:

- Begin Record → action(s)
- Finish Initial Record → action(s)
- Play Clip → action(s)
- Stop Clip → action(s)
- Clear Clip → action(s)

Useful for context-sensitive behaviors (e.g., when a clip plays, stop others in same color).

Targeting / Selection

Direct

Bindings created via MIDI Learn on a specific clip target that clip directly.

Selected Clip

The **Select** action marks a clip (white dot). Other actions targeting "Selected Clip" operate on it. **Select Next/Previous** cycle through clips.

By Color

Actions can target "All clips of [color]", "All playing clips of [color]", etc.

Play Groups

Clips in a play group can be configured as mutex (starting one stops others).

Quantization

Count-In (when recording starts)

Mode	Behavior
None	Immediate
Master	Next master cycle boundary
Loop	If empty: Master. If has audio: next loop cycle
Custom	User-defined interval

Count-Out (when recording stops)

Same modes as Count-In. "Auto Count Out" enables fixed-length recording.

Settings Hierarchy

Project-wide → Color Level → Individual Clip (highest priority)

MIDI Control

No direct "change quantization mode" action. Workarounds:

- Different bindings with different per-action quantization overrides
 - **Set Master Length** changes what "Master" means
 - **Adjust Beats Per Bar** changes time signature
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Clock

- Tempo set by: first recorded loop (auto-detection), tap tempo, manual entry, external sync
 - **MIDI Clock**: LoopyPro sends AND receives. Enable "CLOCK INPUT" for your interface in Settings → MIDI
 - **Ableton Link**: Alternative sync. Don't use both simultaneously.
 - Master Cycle Length is the quantization interval. Adjustable via **Set Master Length** action.
 - Default time signature: 4/4
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State Feedback

LoopyPro sends MIDI feedback to controllers on the **same channel and message type** as the learned trigger.

What generates feedback

- Clip playing/stopped/muted/recording → button LED state
- Parameter values → encoder ring position
- Effect enabled/disabled → button LED state

MFT-specific

- MFT System MIDI Channel must match the control channel
 - Factory default system channel: 4
 - Feedback must be enabled in Control Settings for the controller
 - Disable feedback if strange loops occur
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One-Button Record/Play/Overdub Pattern

A common single-button workflow:

1. **Play/Stop** with "Record If Empty" ON, "After Recording" = Play
2. **Record** with "If Clip Has Audio" = Overdub, timing = "Next Trigger"

Press sequence: record → stop+play → overdub → stop overdub → ...

LoopyPro 2.0 Additions (July 2025)

- MIDI Clips (record/edit/playback MIDI loops, piano roll)
 - Polyphonic Clip Playback (trigger clips as pitched notes or slices via MIDI)
 - Groove Quantization / Warp
 - MIDI Parameter Automation on audio and MIDI loops
 - Latency Compensation for MIDI controllers
 - Absolute timing in actions (seconds.milliseconds)
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Built-In MFT Support

LoopyPro has native Midifighter Twister support:

Automatic Binding Mode (when enabled)

- 4x4 knob grid maps to clips
- LED rings show playhead position
- Button press toggles play/stop
- Colors match track colors

Dial modes: Loop location (default), Volume, Pan, Custom CC

Button behaviors (configurable): None, Toggle Play, Toggle Record, Clear

Manual Binding Mode

Disable automatic bindings → full manual control via MIDI Learn. Retain LED feedback.

Configuration Notes

- Default MFT channel: 2, CC Hold mode, CC 0-15 for bank 1
- System MIDI channel must match for feedback (factory: channel 4)
- Connect in same USB port order each time (iOS can't distinguish identical controllers)
- Disable automatic bindings before adding custom bindings to avoid conflicts